Remember too that an individual's descendants constitute not a single line but a branching line. Whichever of your ancestors it was who 'created' a particular short length of your chromosome 8a, he or she very likely has many other descendants besides you. One of your genetic units may also be present in your second cousin. It may be present in me, and in the Prime Minister, and in your dog, for we all share ancestors if we go back far enough. Also the same small unit might be assembled several times independently by chance: if the unit is small, the coincidence is not too improbable. But even a close relative is unlikely to share a whole chromosome with you. The smaller a genetic unit is, the more likely it is that another individual shares it—the more likely it is to be represented many times over in the world, in the form of copies.

The chance coming together, through crossing-over, of previously existing sub-units is the usual way for a new genetic unit to be formed. Another way—of great evolutionary importance even though it is rare—is called *point mutation*. A point mutation is an error corresponding to a single misprinted letter in a book. It is rare, but clearly the longer a genetic unit is, the more likely it is to be altered by a mutation somewhere along its length.

Another rare kind of mistake or mutation which has important long-term consequences is called *inversion*. A piece of chromosome detaches itself at both ends, turns head over heels, and reattaches itself in the inverted position. In terms of the earlier analogy, this would necessitate some renumbering of pages. Sometimes portions of chromosomes do not simply invert, but become reattached in a completely different part of the chromosome, or even join up with a different chromosome altogether. This corresponds to the transfer of a wad of pages from one volume to another. The importance of this kind of mistake is that, though usually disastrous, it can occasionally lead to the close *linkage* of pieces of genetic material which happen to work well together. Perhaps two cistrons which have a beneficial effect only when they are both present —they complement or reinforce each other in some way—will be brought close to each other by means of inversion. Then natural selection may tend to favour the new 'genetic unit' so formed, and it will spread through the future population. It is possible that gene complexes have, over the years, been extensively rearranged or 'edited' in this kind of way.